

(7.1)(a) Consider the proton to be a uniformly charged sphere of radius 1 fm. Determine the correction to the 1s ground state energy. (4 points)

This is a standard problem. One can calculate the electric field from Gauss' law as in a freshman course and deduce the potential  $\Phi(\vec{r})$  as a function of  $r$ . The energy of the electron is  $-e\Phi(\vec{r})$ . For  $r > a$  the potential is

$$\frac{1}{4\pi\epsilon_0} \frac{e}{r}.$$

For  $r < a$  only the charge inside contributes to the electric field and so it is  $\frac{e(r^3/a^3)}{4\pi\epsilon_0 r^2} = \frac{1}{4\pi\epsilon_0} \frac{r}{a^3}$  and so the potential contains a contribution

$$-\frac{1}{4\pi\epsilon_0} e \frac{r^2}{2a^3}$$

apart from an additive contribution chosen to make the potential continuous. So we have for  $r < a$

$$V(r) = \frac{1}{4\pi\epsilon_0} e \left( -\frac{r^2}{2a^3} + \frac{3}{2a} \right)$$

The contribution to the Hamiltonian is  $-e\Phi(r)$  and this differs from the unperturbed Hamiltonian **only** for  $r < a$ . The difference is the perturbing Hamiltonian and is given by

$$H' = -\frac{q^2}{a} \left( -\frac{r^2}{2a^2} + \frac{3}{2} - \frac{a}{r} \right)$$

for  $r < a$  and vanishes for  $r > a$ . Now we need to evaluate  $\langle 1s|H'|1s\rangle$ . Since the nucleus is much smaller than the Bohr radius it makes eminent sense to approximate  $|\psi_{1s}(r)|^2$  by  $|\psi_{1s}(0)|^2$  which is given by  $\frac{1}{\pi a_B^3}$  where  $a_B$  is the Bohr radius. Since the proton is not a spherical distribution with a sharp edge doing the integral carefully is unwarranted. We just need

$$4\pi \int_0^a dr r^2 H' = -\frac{q^2}{a} \times \left( -\frac{2\pi a^3}{5} \right) = \frac{2\pi q^2 a^2}{5}.$$

Thus we obtain

$$E^{(1)} = \frac{q^2}{2a_B} \frac{4}{5} \left( \frac{a}{a_B} \right)^2 \approx 3.1 \times 10^{-9} \text{ eV} \approx 3.87 \times 10^{-9} \text{ eV}.$$

This is approximately 936 kHz.

This is not a theoretical problem only. One can calculate the effect of the nuclear radius on the 2S state and check how the difference in energy between the 2S and 1S states is affected by this term. Indeed it has been used to provide an estimate of the charge radius

of the proton! These incredibly precise experiments were conducted by the group of Ted Hänsch who shared the 2005 Nobel Prize in Physics for precision spectroscopy. The  $2S - 1S$  transition is especially slow as we will see next quarter.

*(b) Consider the nuclear shell model. Neutrons and protons are treated as independent particles in a three-dimensional isotropic harmonic oscillator potential. Each nucleon has spin-orbit coupling  $-2a\vec{\ell} \cdot \vec{s}$  with  $a > 0$  that is a perturbation on the confining harmonic oscillator potential. Write down the three lowest energy single-particle states by specifying the correct quantum numbers. You are not asked to write down the wave functions. Determine the ground state angular momentum (and parity if you can) of (i) tritium  ${}^3\text{H}_1$ , (ii)  ${}^{11}\text{B}_5$ . (5 points)*

The energy levels of the three-dimensional oscillator can be found on page 352 of Shankar. The lowest energy state has  $n\ell = 0$ . The first excited state is characterized by  $n = \ell = 1$  and is three-fold degenerate. The next state has  $n = 2$  and  $\ell = 0$  or  $2$  and this has degeneracy of 6.

Since  $a > 0$  the states in which  $\vec{\ell}$  and  $\vec{s}$  are parallel have the lowest energy. The ground state has zero orbital angular momentum and no spin-orbit effects. It can be denoted by  $S_{1/2}$ . The next state splits into a  $P_{3/2}$  and a  $P_{1/2}$  level with the  $P_{3/2}$  level being lower in energy.

$$H_{so} = -a[j(j+1) - \ell(\ell+1) - s(s+1)]$$

which has the value  $-a[(15/4) - 2 - (3/4)] = -a$  for  $P_{3/2}$  and  $-a \times [(3/4) - 2 - (3/4)] = 2a$  for  $P_{1/2}$ . (What is the center of gravity of these six states?) Thus the lowest three levels are  $S_{1/2}$  with energy  $(3/2)\hbar\omega$ ,  $P_{3/2}$  with energy  $(5/2)\hbar\omega - a$  with degeneracy 4 and then the  $P_{1/2}$  state with energy  $(5/2)\hbar\omega + 2a$  and degeneracy 2.

In tritium the two neutrons occupy the  $S_{1/2}$  state and yield a net angular momentum of zero. The one proton occupies the  $S_{1/2}$  state and so  $J = 1/2$  for this state.

The six neutrons completely fill the  $S_{1/2}$  and  $P_{3/2}$  shells with a net angular momentum zero and the 5 protons fill the same two shells leaving a “hole” behind. Thus, the angular momentum is  $J = 3/2$ .

If you are a nucleophile you should figure out the parity of these states.

Maria Goeppert-Mayer’s Nobel lecture is a nice readable introduction to magic numbers in nuclei and the nuclear shell model.

c) The dissociation energy of  $H_2$  is 4.46 eV. You are given that the dissociation energy of the  $D_2$  molecule is 4.54 eV. In the simplest approximation would you expect any difference in the dissociation energies for the two molecules. Please justify your answer. Now consider the vibrational motion of the nuclei and the associated zero-point energy to explain the difference. Find the zero-point energy of the  $H_2$  molecule. (5 points)

If one treats the nuclei as infinitely massive in computing the energy of the molecule (in the ground state we compute  $\epsilon_+(R)$  the symmetric state for the two electrons) only the charge enters the Hamiltonian and therefore, both  $H_2$  and  $D_2$  yield the same dissociation energy. In the adiabatic approximation we can interpret  $\epsilon_+(R)$  as the potential energy of interaction between the two nuclei. Expanding around the minimum of the potential that is assumed to occur at  $R_0$  we have

$$\epsilon_+(R) \approx \epsilon_+(R_0) + \frac{k}{2}(R - R_0)^2.$$

Thus the spring constant is the same for both the molecules. The frequency of the associated oscillator will differ because of the difference in masses.

What is the energy of the diatomic molecule? The state is specified by  $n$  the excitation of the one-dimensional harmonic oscillator corresponding to the vibrational motion and  $J$  the rotational quantum number:

$$E = \epsilon_+(R_0) + \hbar\omega \left( n + \frac{1}{2} \right) + BJ(J+1) + \text{higher order terms}.$$

In the ground state  $n = 0$  and  $J = 0$ . The dissociation energy is the difference in energy between the ground state energy of the molecule and that of two non-interacting atoms ( $R \rightarrow \infty$ ) taken to be zero. Thus the dissociation energy is given by

$$E_{dis} = -\epsilon_+(R_0) - \frac{1}{2}\hbar\omega.$$

$\epsilon_+(R_0) < 0$  and so the dissociation energy is positive. It is reduced by the zero-point fluctuations. Since  $\epsilon_+(R_0)$  is the same for the hydrogen and deuterium molecules, we have

$$E_{dis,D} - E_{dis,H} = \frac{1}{2}(\hbar\omega_H - \hbar\omega_D).$$

Since  $\omega = \sqrt{k/\mu}$  where  $k$  is the spring constant and  $\mu$  is the reduced mass and  $k$  is the same for both the molecules we have

$$\frac{\omega_D}{\omega_H} \approx \frac{1}{\sqrt{2}}.$$

Thus, given the difference in dissociation energies to be  $0.08\text{ eV}$  we obtain the zero-point energy of hydrogen:

$$\frac{1}{2}\hbar\omega_H\left(1 - \frac{1}{\sqrt{2}}\right) = 0.08\text{ eV} \Rightarrow \frac{1}{2}\hbar\omega_H = 0.27\text{ eV}.$$

**17.2.2.** The unperturbed Hamiltonian is  $-\gamma B_0 S_z$  with eigenvectors  $|+\rangle$  and  $|-\rangle$  with energies  $E_+^0 = -\gamma B_0 \hbar/2$  and  $E_-^0 = \gamma B_0 \hbar/2$  respectively using obvious notation.

The perturbation is  $H' = -\gamma B \sigma_x/2$ . The first-order corrections  $\langle +|\sigma_x|+\rangle$  and  $\langle -|\sigma_x|-\rangle$  vanish. The second-order corrections are given by

$$E_+^2 = \frac{|\langle -|H'|+\rangle|^2}{E_+^0 - E_-^0} = -\frac{\gamma \hbar B^2}{4B_0}.$$

It is easy to see that

$$E_-^2 = \frac{\gamma \hbar B^2}{4B_0}.$$

The corrections to the eigenstates are easily calculated:

$$|+^1\rangle = |-\rangle \frac{\langle -|H'|+\rangle}{E_+^0 - E_-^0} = \frac{B}{2B_0} |-\rangle.$$

$$|-^1\rangle = |+\rangle \frac{\langle +|H'|-\rangle}{E_-^0 - E_+^0} = -\frac{B}{2B_0} |+\rangle.$$

It is easy to find the exact eigenvalues and eigenvectors by diagonalizing

$$H = -\frac{\gamma \hbar}{2} \begin{pmatrix} B_0 & B \\ B & -B_0 \end{pmatrix}.$$

So the eigenvalues are

$$\Lambda_{\pm} = \mp \frac{\gamma \hbar}{2} \sqrt{B_0^2 + B^2}.$$

Expansion yields

$$\mp \frac{\gamma \hbar}{2} \left( B_0 + \frac{B^2}{2B_0} \cdots \right).$$

The unnormalized eigenvector for the ground state can be written down by inspection:

$$\begin{pmatrix} \frac{1}{\sqrt{B_0^2 + B^2} - B_0} \\ \frac{1}{B} \end{pmatrix} \approx \begin{pmatrix} \frac{1}{2B_0} \\ \frac{B}{2B_0} \end{pmatrix}.$$

This agrees with the perturbation result. The other eigenvector yields

$$\begin{pmatrix} -\frac{B}{B_0 + \sqrt{B_0^2 + B^2}} \\ 1 \end{pmatrix} \approx \begin{pmatrix} -\frac{B}{2B_0} \\ 1 \end{pmatrix}.$$

**17.2.4** The Thomas-Reiche-Kuhn sum rule is not only useful in spectroscopy (we will discuss oscillator strengths next quarter) but has inspired many sum rules in condensed

matter and particle physics. One usually starts from a commutator of operators or currents. In our case, the usual starting point would be

$$-i\langle n|[X, P]|n\rangle = 1$$

use completeness and  $P = im[H, X]$  to obtain the sum rule. We start from the given expression.

We write down the following obvious steps:

$$\begin{aligned} \sum_{n'} (E_{n'} - E_n) |\langle n'|X|n\rangle|^2 &= \sum_{n'} (E_{n'} - E_n) \langle n|X|n'\rangle \langle n'|X|n\rangle \\ &= \sum_{n'} [\langle n|XH|n'\rangle \langle n'|X|n\rangle - \langle n|HX|n'\rangle \langle n'|X|n\rangle] \end{aligned}$$

where we have used  $\langle n|H = \langle n|E_n$  etc. Completeness of the set  $\{|n'\rangle\}$  allows one to simplify the right-hand side:

$$\sum_{n'} (E_{n'} - E_n) |\langle n'|X|n\rangle|^2 = \langle n|XHX - HX^2|n\rangle. \quad (1)$$

Since  $\langle n|HX^2|n\rangle = E_n \langle n|X^2|n\rangle = \langle n|X^2H|n\rangle$  we can write

$$\sum_{n'} (E_{n'} - E_n) |\langle n'|X|n\rangle|^2 = \frac{1}{2} \langle n|2XHX - HX^2 - X^2H|n\rangle.$$

Now we note that

$$2XHX - HX^2 - X^2H = [X, [H, X]]$$

as is verified by expansion. Using

$$[H, X] = \left[ \frac{P^2}{2m}, X \right] = -\frac{i\hbar}{m} P$$

we obtain

$$[X, [H, X]] = -\frac{i\hbar}{m} [X, P] = \frac{\hbar^2}{2m}.$$

This yields the celebrated sum rule.

The other part is elementary algebra using

$$X = \sqrt{\frac{\hbar}{2m\omega}} (a + a^\dagger) \quad \text{and} \quad \langle n'|X|n\rangle = \sqrt{\frac{\hbar}{2m\omega}} [\sqrt{n} \delta_{n',n-1} + \sqrt{n+1} \delta_{n',n+1}].$$

Thus we obtain

$$|\langle n'|X|n\rangle|^2 = \frac{\hbar}{2m\omega} [n \delta_{n',n-1} + (n+1) \delta_{n',n+1}].$$

For the first term on the right-hand side above we have  $E_{n'} - E_n = -\hbar\omega$  and from the second term  $E_{n'} - E_n = \hbar\omega$ . Thus we obtain

$$\sum_{n'} (E_{n'} - E_n) |\langle n' | X | n \rangle|^2 = \frac{\hbar}{2m\omega} \hbar\omega (-n + (n+1)) = \frac{\hbar^2}{2m}.$$

**17.2.7. Caveat:** *I might have pulled a fast one over you by doing an even number of sign errors or canceling errors since I knew the answer.*

The algebra involved in the problem is trivial once you guess  $\Omega$ . I wanted to make sure you read *Method 2* on page 462 and understood this clever technique.

$H^1 = -qfX$ . Since we need  $[\Omega, H^0] \propto X$  it is natural to guess  $\Omega = AP$  and determine  $A$ . We have

$$[AP, H^0] = \frac{1}{2}m\omega^2 A [P, X^2] = -i\hbar m\omega^2 A X.$$

So we choose  $A = -i\frac{qf}{\hbar m\omega^2}$  to obtain  $[\Omega, H^0] = H^1$ .

Now that we have  $\Omega$  we need the commutators in 17.2.39. Since this is a simple harmonic oscillator we employ creation and annihilation operators. Using

$$X = \sqrt{\frac{\hbar}{2m\omega}} (a + a^\dagger) \quad \text{and} \quad P = i\sqrt{\frac{m\hbar\omega}{2}} (a^\dagger - a)$$

we obtain the different matrix elements:

$$\langle n^0 | \Omega | n^0 \rangle \propto \langle n^0 | P | n^0 \rangle = 0.$$

$$\langle n^0 | H^1 \Omega | n^0 \rangle = -\frac{qf^2}{2m\omega^2} \langle n^0 | (a^\dagger + a)(a^\dagger - a) | n^0 \rangle = -\frac{qf^2}{2m\omega^2}.$$

Substituting into Equation 17.2.39 the second term vanishes and we obtain the same result as in equation 17.2.6.

**17.3.1** The dipole selection rule according to Shankar states that  $\ell' = \ell \pm 1$  and thus in the Hamiltonian for the Stark effect for the  $n = 2$  states of hydrogen all the matrix elements with  $\ell = \ell'$  vanish. So only matrix elements between  $2s$  and  $2p$  states can be non-vanishing.

From 17.2.21 we also need  $\Delta m = 0$  for the matrix elements of  $Z$ . Thus the only independent, possibly (and actually!) non-vanishing matrix element is  $\langle 200|H'|210\rangle$ . This determines its Hermitian conjugate, of course.

We need

$$\langle 200|e\mathcal{E}r\cos\theta|210\rangle = \int_0^{2\pi} d\phi \int_0^\pi d\theta \sin\theta Y_{00}^* \cos\theta Y_{10} \int_0^\infty dr r^2 R_{20}(r) r R_{21}(r)$$

where all the functions are known and the integrals can be evaluated. The wave functions written as a product of  $R_{n\ell}(r)$  and  $Y_\ell^m(\theta, \phi)$  are given by

$$\psi_{2,0,0} = \left(\frac{1}{2a_0^3}\right)^{1/2} \left(1 - \frac{r}{2a_0}\right) e^{-\frac{r}{2a_0}} \times \frac{1}{\sqrt{4\pi}} \quad (2)$$

$$\psi_{2,1,0} = \left(\frac{1}{24a_0^3}\right)^{1/2} \frac{r}{a_0} e^{-\frac{r}{2a_0}} \times \sqrt{\frac{3}{4\pi}} \cos\theta \quad (3)$$

Let us assume you wimp out and go to *Mathematica*. The radial integral are easily evaluated in units of  $a_0$ ,  $u \equiv r/a_0$  ( $H' = e\mathcal{E}a \times u \cos\theta$ ):

```
In[8]:= R20[u_]:= (1/Sqrt[2]) (1 - u/2) Exp[-u/2]
In[9]:= R21[u_]:= Sqrt[1/24] u Exp[-u/2]
In[10]:= Integrate[R20[u] u^3 R21[u], {u,0,Infinity}]
Out[10]= -3 Sqrt[3]
```

The angular integral, including a factor of  $2\pi$  for the  $\phi$  integral

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In[14]:= Y10[theta_]:= Sqrt[3/(4 Pi)] Cos[theta]
In[15]:= Y00[theta_]:= 1/Sqrt[4 Pi]
In[18]:= Integrate[2 Pi Sin[theta] Y00[theta] Cos[theta] Y10[theta], {theta, 0, Pi}]
Out[18]= 1/Sqrt[3]
```

Multiplying the values of the two integrals and restoring the factor  $e\mathcal{E}a$  we obtain

$$\langle 200|e\mathcal{E}r\cos\theta|210\rangle = -3e\mathcal{E}a_0. \quad (4)$$

The Hamiltonian  $H'$  within this subspace is given by

$$\hat{H}' = \begin{bmatrix} 0 & -3e\mathcal{E}a_0 & 0 & 0 \\ -3e\mathcal{E}a_0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (5)$$



So the state  $|211\rangle$  and  $|21-1\rangle$  are unaffected while  $|200\rangle$  and  $|210\rangle$  are mixed; we can diagonalize  $H'$  in this subspace and find the corrections to the eigenvalue  $E_2^0$  as

$$E_2^{(1)} = \pm 3e\mathcal{E}a_0. \quad (6)$$

The eigenvectors are

$$\frac{1}{\sqrt{2}} [ |200\rangle \pm |210\rangle ]. \quad (7)$$

Thus the fourfold degeneracy is partially lifted and there is an energy change *linear* in  $\mathcal{E}$ . Note that the splitting could be guessed in order of magnitude as  $e\mathcal{E}a_0$  on dimensional grounds.